AI Engineer

MID-LEVEL ROLE PROFILE | Artificial Intelligence & Machine Learning

Position overview

Designs, builds, evaluates, and deploys AI/ML solutions that address business needs. Works with product, data, engineering, and operations teams to turn data and prototypes into reliable, maintainable systems.

Experience and education

- Typically 3–5 years of relevant experience in software engineering, machine learning, data science, or applied AI, including hands-on delivery of at least one model or AI-enabled application into production.
- Bachelor's degree in Computer Science, Engineering, Mathematics, Statistics, or a related field; equivalent practical experience may be accepted. A postgraduate degree can be an advantage, but is not universally required.

Required skills

- Strong Python programming; working knowledge of SQL, Git, testing, debugging, and clean-code practices.
- Machine-learning fundamentals: supervised and unsupervised learning, feature engineering, model selection, validation, and error analysis.
- Experience with common libraries such as scikit-learn and at least one deep-learning framework such as PyTorch or TensorFlow.
- Data preparation and evaluation: pandas/NumPy, data quality checks, metrics, experiment tracking, and reproducible workflows.
- Production basics: REST APIs, containers, CI/CD, cloud services (AWS, Azure, or GCP), monitoring, and model/version management.
- For generative AI roles: practical familiarity with LLM APIs, embeddings, retrieval-augmented generation (RAG), vector search, prompt evaluation, and safeguards.
- Communication skills to explain model trade-offs, limitations, risks, and results to technical and non-technical stakeholders.

Key responsibilities

- Translate business problems into measurable AI use cases, baselines, and evaluation criteria.
- Prepare datasets, develop features, train or adapt models, and compare approaches using appropriate metrics.
- Build prototypes and production services; collaborate with software engineers on integration and deployment.
- Monitor model quality, latency, cost, drift, and reliability; investigate issues and improve performance.
- Document datasets, experiments, assumptions, limitations, and operational procedures.
- Apply privacy, security, fairness, explainability, and human-review practices appropriate to the use case.

Preferred qualifications

- Experience operating ML systems in production and using MLOps or experiment-tracking tools.
- Domain experience relevant to the employer, such as NLP, computer vision, forecasting, recommendations, or conversational AI.
- Portfolio, technical project, or open-source contribution demonstrating end-to-end delivery.

What success looks like

- Models and AI features meet agreed quality, latency, cost, and reliability targets.
- Solutions are maintainable, monitored, documented, and adopted by intended users.
- Experiments produce clear evidence for decisions, including when not to use a model.

Experience ranges and tool lists are indicative hiring guidelines, not universal requirements. Tailor this profile to the product, domain, and level of production ownership.